

Two sample statistics

Exercise 1. The medical school of Wisconsin conducted an investigation to determine whether women and men complete medical school in significantly different amounts of time, on the average. Two independent random samples were selected and the following summary information concerning times to completion of medical school computed:

	Women	Men
Sample Size	80	100
Sample mean	4.5 years	4.4 years
Population Standard Deviation	0.5 years	0.6 years

1. State the appropriate null and alternative hypotheses
2. Perform the appropriate test of hypothesis to determine whether there is a significant difference in time to completion of medical school between women and men. Test using $\alpha = 0.05$.
3. Find the p -value associated with the test in part b and explain how to use it for testing the hypothesis in part a.

Exercise 2. According to a recent survey, women chat on their mobile phones more than men (CNN.com, August 25, 2010). Irina wants to estimate the difference between the average chat time for female and male college students. She takes a random sample of 40 male students and 40 female students in her college. She finds that female students chatted for a sample average of 820 minutes per month with a sample standard deviation of 160 minutes. Male students, on the other hand, chatted for a sample average of 760 minutes per month with a sample standard deviation of 240 minutes. You may assume that the population variances are equal.

1. Construct a 95% confidence interval for the difference between the population means.
2. Can we conclude that average chat time is different between female and male college students with 95% confidence?

Exercise 3. A student has to decide whether he wants to major in electrical engineering or computer science. To find out which job pays more, he interviews eight alumni in each profession and asks them what their starting salary was. The electrical engineering alumni had a mean starting salary of \$54,000 and a standard deviation of \$8,000. The computer science alumni had a mean starting salary of \$62,000 and a standard deviation of \$10,000. It is reasonable to assume equal population variances, and assume starting salaries are normally distributed.

- a) Specify the competing hypotheses to determine if electrical engineering and computer science majors mean starting salaries are equal.
- b) Calculate the value of the test statistic, and find the critical value at a 5% significance level.
- c) Make a conclusion at the 5% significance level.

Exercise 4. A 7000-seat theatre is interested in determining whether there is a difference in attendance between shows on Tuesday evening and those on Wednesday evening. Two independent samples of 25 weeks are collected for Tuesday and Wednesday. The mean attendance on Tuesday evening is calculated as 5500, while the mean attendance on Wednesday evening is calculated as 5850. The known population standard deviation for attendance on Tuesday evening is 550 and the known population standard deviation for attendance on Wednesday evening is 445.

- a) What are the appropriate hypotheses to determine whether there is a difference in attendance for shows on Tuesday evening and Wednesday evening. Use Tuesday attendance as population 1 and Wednesday attendance as population 2 or μ_D as the mean difference in matched-pairs sampling.
- b) Compute the value of the appropriate test statistic.
- c) What is the appropriate decision given a 5% level of significance?

Exercise 5. 2022/23 exam Q1 (modified)

A researcher is interested in the idea of a Protestant work ethic. She conducts a random survey on weekly hours worked by people in employment. The 3,453 Protestants in the survey work a mean of 39.360 hours per week, with a standard deviation of 13.182 hours, while the 6,044 Catholics work a mean of 42.206 hours per week with a standard deviation of 13.690 hours.

- a) Assuming the population variances are the same is there any evidence that Protestants work more hours than Catholics?
- b) Redo part a assuming the population variances are different.
- c) Does this tell us whether the Protestant work ethic exists? Be sure to explain why or why not.

Exercise 6. Two different teaching methods are being compared for effectiveness in a statistics course. Method A allowed students to use computers during instruction and assessments, while Method B required students to use only pen and paper. Method A was used with 45 students who scored a mean of 76.4 on the final exam with a standard deviation of 12.5. Method B (pen and paper only) was used with 50 students who scored a mean of 81.2 on the same final exam with a standard deviation of 14.8. You can assume that the exam scores are normally distributed.

- Construct a 90% confidence interval for the difference in mean exam scores between the pen-and-paper method and the computer method.
- Is there statistical evidence that the pen-and-paper method (Method B) produces higher exam scores than the computer method (Method A)? Test at the 5% significance level.
- The department chair wants to know if the superiority of the pen-and-paper method exceeds 5 points. Formulate and test the appropriate hypotheses.

Exercise 7. A pharmaceutical company is testing the effectiveness of a new medication for reducing cholesterol levels. They conduct a study with 60 patients who receive the medication and 60 patients who receive a placebo. After 30 days, the treatment group shows a mean reduction in cholesterol of 42.5 mg/dL with a standard deviation of 8.3 mg/dL. The placebo group shows a mean reduction of 38.9 mg/dL with a standard deviation of 7.6 mg/dL. Assume the population variances are equal.

- State the appropriate null and alternative hypotheses to determine if the medication is effective at reducing cholesterol.
- Calculate the test statistic and determine the critical value at a 1% significance level.
- Based on your results, would you recommend this medication for cholesterol reduction? Justify your answer.

Exercise 8. An investment advisor claims that stocks in technology companies have shown higher annual returns compared to stocks in energy companies over

the past five years. To verify this claim, she randomly samples 12 technology stocks and 15 energy stocks. The technology stocks had a mean annual return of 11.2% with a standard deviation of 4.8%, while the energy stocks had a mean annual return of 8.6% with a standard deviation of 5.2%. Assume the returns are normally distributed.

- Set up the null and alternative hypotheses to test the advisor's claim.
- Calculate the pooled variance estimate and the value of the test statistic.
- Determine the p -value of your test and make a conclusion at the 5% significance level.
- Calculate a 95% confidence interval for the difference in mean annual returns between technology and energy stocks.

Exercise 9. A city health department wants to determine if a new public health campaign has been effective at increasing the rate of flu vaccination among seniors. Before the campaign, approximately 65% of seniors reported getting the flu vaccine. After the campaign, the health department surveyed 400 randomly selected seniors and found that 278 had received the flu vaccine. A neighboring city with similar demographics but without the campaign surveyed 350 seniors and found that the vaccination rate was 68% (238 out of 350).

- Formulate the null and alternative hypotheses to test whether the vaccination campaign significantly increased the vaccination rate compared to the neighboring city.
- Calculate the test statistic for comparing these two proportions.
- At a 5% significance level, determine if the vaccination campaign was effective.
- Construct a 95% confidence interval for the difference in vaccination rates between the two cities.